CausalVerify: An Execution-Grounded Benchmark for LLM Causal Inference Workflows

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Abstract

Existing causal-inference benchmarks for LLMs often evaluate text descriptions of methods or whether generated code can run. They rarely test whether the executed workflow recovers the target causal estimate. CausalVerify studies this verification problem by separating realistic interpretation from verifiable computation. The benchmark draws on 259 published economics papers, each represented by a reconstructed research question, data description, and institutional context, together with 100 fixed-seed synthetic scenarios that realise CSV datasets for difference-in-differences (DID), event study (ES), instrumental variables (IV), and regression discontinuity design (RDD). Experiment A scores method-family and direction agreement against four-LLM consensus labels. Experiment B runs model-written R code and checks whether the extracted treatment-effect estimate matches a canonical estimator on the same realised dataset. A calibration arm asks whether self-reported confidence separates correct from incorrect workflows. On Experiment B, seven evaluated LLMs reach final L2b+ pass rates between 10% and 88%. Execution ranking (L2b) agrees with coefficient recovery (L2b+) far better than text-direction scoring (L4): Kendall $\tau = 0.81$ and Spearman $\rho = 0.93$, versus Kendall τ between -0.20 and 0.10 for L4. A robustness run with Llama-3.3-70B-Instruct shows the same qualitative gap, and reported confidence does not reliably separate correct from incorrect workflows. Code, data, cached outputs, and a datasheet are released with the submission.

1 Introduction

Evaluating LLMs for empirical causal inference is a verification problem, not merely a text-scoring problem. A valid empirical claim requires an estimand, an identification argument, an estimator, executable analysis code, and a numerical result whose interpretation matches the target causal quantity [49, 47, 20, 46, 24, 3]. This structure exposes a failure mode that standard LLM evaluations can miss: a model may name the right empirical design, produce code that runs without error, and still estimate the wrong coefficient. In applied research, such failures are more dangerous than obvious failures because the response appears fluent, complete, and confident while the causal claim remains unverified. This concern sits within broader verification pressure in scientific publishing, where more than 10,000 papers were retracted in 2023, a record high reported by Van Noorden

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Track on Evaluations and Datasets.

[57]. As LLMs move from text assistance toward software engineering and scientific-analysis agents [14, 61, 52, 39, 58, 21, 27, 43, 12], causal-inference evaluation must ask not only whether a model sounds right, but whether its executed workflow recovers the right estimate.

Existing benchmarks capture only parts of this problem. Prior work and benchmarks on LLM causal reasoning study causal direction, graph reasoning, counterfactuals, correlation-to-causation judgments, intervention design, surveys of benchmark validity, and real-world identification tasks [31, 62, 29, 28, 11, 13, 60, 41, 63, 2, 35, 53, 51]. Code benchmarks evaluate whether programs can be synthesized, executed, or made to pass tests [10, 5, 19, 36, 40, 9, 33, 17, 27, 25, 64]. These settings are valuable, but causal workflows require a stricter check: after a model interprets a study, chooses a design, writes analysis code, and reports a treatment effect, does the executed workflow recover the intended causal estimate? This gap matters because code can pass surface checks while remaining semantically wrong [38].

We introduce CAUSALVERIFY, an execution-grounded benchmark for LLM causal-inference workflows. The benchmark deliberately separates ecological realism from executable verifiability. Real published papers provide realistic institutional context, but rarely provide the cleaned data, final runnable specification, and counterfactual treatment effect needed for direct verification. We therefore use 259 published economics papers to measure text-level agreement on method family and effect direction, not to certify causal correctness. Conversely, fixed-seed synthetic DGPs sacrifice some realism but provide executable targets: 100 realised CSV datasets across difference-in-differences, event study, instrumental variables, and regression discontinuity designs [22, 4, 55, 23, 34, 8, 42, 6, 1, 7, 16, 54, 50, 48]. Models write R code, the benchmark executes it, and L2b+ checks whether the extracted treatment-effect estimate matches a canonical estimator on the same realised dataset.

CAUSALVERIFY separates four signals that are often conflated: textual interpretation, code executability, numerical coefficient recovery, and self-assessed confidence. Across seven evaluated LLMs, final L2b+ pass rates range from 10% to 88%. Execution success is strongly aligned with coefficient recovery (Kendall $\tau = 0.81$, Spearman $\rho = 0.93$), whereas text-direction scoring is weakly or negatively aligned with L2b+ ranking (Kendall $\tau \in [-0.20, 0.10]$). Execution is not sufficient—some code runs and estimates the wrong coefficient—but it is more informative than plausible directional text. A calibration arm further shows that retrospective confidence does not cleanly separate correct from incorrect workflows, connecting the benchmark to work on calibration, verbalized uncertainty, selective prediction, and self-checking [18, 15, 26, 37, 45, 30, 32, 44, 56, 59]. Our contributions are a benchmark design separating real-paper agreement from executable correctness, L2b+ as an execution-grounded coefficient-recovery metric, and empirical evidence that plausible text, runnable code, numerical correctness, and confidence can diverge sharply.

2 Related Work and Positioning

Prior work evaluates important pieces of the LLM causal-analysis workflow, but usually not the executed causal estimate. Causal benchmarks test textual reasoning about direction, counterfactuals, graphs, interventions, or identification. Code benchmarks test whether generated programs execute or pass functional tests. Science-agent benchmarks evaluate broader discovery or tool-use workflows. CAUSALVERIFY differs in its verification target: the primary correctness signal is not a text label or a generic program test, but whether executed model-written analysis code recovers the target treatment-effect estimate.

The closest prior benchmark is CauSciBench [2], which evaluates end-to-end causal-analysis performance over curated real, synthetic, and textbook tasks. CAUSALVERIFY asks a complementary measurement question: when a model appears to understand a causal task, which layer of the workflow is actually correct? It separates real-paper text agreement, controlled-DGP execution verification, and retrospective calibration rather than collapsing them into a single causal-ability score.

The closest concurrent work is Sawarni et al. [51], which disentangles causal identification from estimation on real-world datasets. Its finding that strategy identification is easier than full specification is consistent with our name-versus-compute gap. The methodological difference is the unit of verification: CAUSALVERIFY requires models to generate and execute R code on controlled DGP data, then checks the recovered coefficient against a realised-data canonical estimate. This isolates

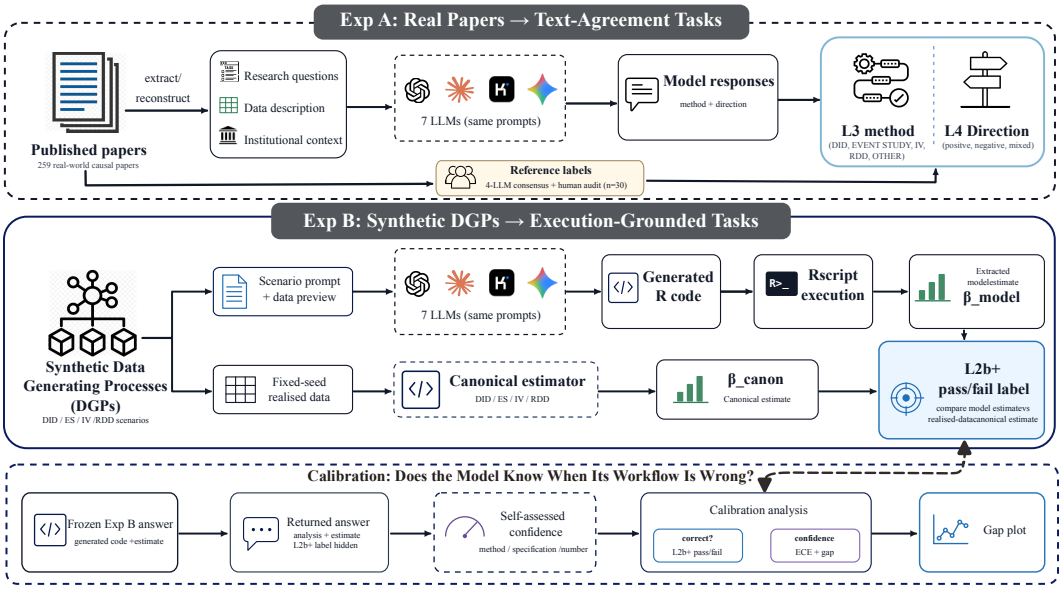


Figure 1: CAUSALVERIFY benchmark overview. Exp A uses published economics papers to form text-agreement tasks for method family and direction. Exp B uses fixed-seed synthetic DGPs and realised datasets to evaluate executable model-written R workflows against canonical estimators. The calibration arm returns each model its own Exp B answer and compares self-assessed confidence with L2b+ correctness.

the failure mode where code runs but computes the wrong causal quantity. Table 1 summarizes the distinction from adjacent benchmarks.

Table 1: Comparison with adjacent benchmarks. ✓ = primary focus; ∼ = partial; ✗ = absent. CAUSALVERIFY combines real-paper context, execution-grounded coefficient verification, layered diagnostics, and self-assessment in one frozen evaluation package.

Benchmark	N	Target	Real ctx.	Exec. coeff.	Layered Dx	Self-assess.
CLadder / CORR2CAUSE	11k / 400	Causal QA / direction	∼	✗	✗	✗
CausalBench	3	Graph + effect	✗	✗	∼	✗
EconCausal	750	Economic causal sign	✓	✗	✗	✗
CauSciBench	367	End-to-end CI pipeline	✓	∼	∼	✗
InterveneBench	744	Intervention design	✓	∼	∼	✗
CausalReasoningBench	173	ID + estimate	✓	✗	∼	✗
Code benchmarks	–	Program tests	✗	∼	∼	✗
Discovery / science agents	264 / 102	Discovery workflow	✓	∼	∼	✗
CAUSALVERIFY	259 + 100	Causal workflow	✓	✓	✓	✓

3 Benchmark Construction and Controls

CAUSALVERIFY separates three evaluation regimes because no single setting provides both ecological realism and executable truth. Real papers test whether models parse applied research contexts, but their ground truth is interpretive: the original cleaned data, exact runnable specification, and counterfactual effect are usually unavailable. This reflects the structure of empirical causal work, where claims depend on the alignment of estimand, identification, estimator, and data [3, 24]. Synthetic DGPs give up some realism but make coefficient correctness executable. Calibration then asks the deployment-facing question: when no external verifier is available, can the model's own confidence serve as a useful warning signal?

Exp A: real-paper context. Exp A contains 259 retained published economics papers after excluding two source failures. Each paper is represented by three reconstructed fields: research question, data description, and institutional context. Seven models are run on every paper, for 1813 outputs. Exp A is intentionally a text-level benchmark, comparable to prior LLM causal-reasoning evaluations that score textual causal direction, method recognition, sign judgments, or identification/estimation judgments [29, 28, 35, 51]. It measures method-family agreement (L3) and direction agreement (L4) against four-LLM consensus reference labels. The benchmark does not treat these labels as human-gold causal truth; instead, it uses real papers to test whether models read applied study contexts in the same way as a structured consensus procedure. Verified causal correctness is reserved for Exp B.

Exp B: synthetic DGP execution. Exp B contains 100 fixed-seed synthetic DGP scenarios across standard applied causal-inference designs: difference-in-differences and event studies, instrumental variables, and regression discontinuity designs [7, 54, 42, 22, 4, 55, 23, 34]. The 100 scenarios comprise 30 DID, 24 Event Study, 24 IV, and 22 RDD tasks; the realised datasets and canonical estimates are fixed before model evaluation. Models receive the research question, data description, and a preview of the realised CSV, then produce runnable R code. We execute the code and compare the extracted treatment-effect estimate to a canonical estimator computed on the same realised dataset. This realised-data comparison avoids penalizing models for finite-sample deviations between the realised sample and the ideal DGP parameter.

Calibration: self-assessment. We attempt a calibration query for each frozen Exp B output; response availability yields 646 calibration records. The model is shown its own previous analysis without ground truth and asked to report confidence in the method choice, specification, and numerical estimate. This is a retrospective verbalized-confidence protocol rather than token-probability calibration, connecting the benchmark to prior work on neural-network calibration, LLM self-knowledge, and confidence elicitation [18, 30, 56, 59]. The resulting confidence is compared to L2b+ correctness. This protocol deliberately mimics deployment: a user may see a model's confidence even when no canonical estimator is available.

Controls and claim boundaries. The benchmark uses a multi-stage evaluation pipeline rather than an autonomous causal scientist. Separate stages construct contexts, audit leakage, collect model responses, extract coefficients, aggregate labels, and elicit calibration, reducing "task writer = respondent = grader" circularity. The synthetic scenarios are not free-form LLM inventions: mathematical templates, treatment-effect signs, sample sizes, seeds, realised CSV files, and canonical estimators are fixed by the benchmark. Natural-language descriptions can vary, but a model cannot pass L2b+ by guessing the sign or recognizing a phrase; it must produce code whose estimated coefficient agrees with the realised-data reference. These components support different claims: Exp A is realistic but text-derived, Exp B is executable but synthetic, and calibration measures confidence–correctness alignment. The benchmark is strongest when these components are read together rather than collapsed into a single leaderboard.

4 Evaluation Layers

The benchmark uses six scoring labels, but they apply to different regimes. L1–L2b+ are Exp B workflow-implementation layers, while L3/L4 are Exp A text-agreement diagnostics. L1 checks that an output exists. L2a checks for a parseable R code block. L2b executes the code with `Rscript` in the fixed evaluation environment. L2b+ is the execution-grounded correctness layer: it checks whether the extracted treatment-effect estimate matches the realised-data canonical estimate within the default relative-error tolerance. L3 measures method-family agreement and L4 measures direction agreement against consensus reference labels. Only L2b+ is a numerical correctness layer; L1/L2a are bookkeeping diagnostics, L2b is execution, and L3/L4 are text agreement.

Why L2b+ uses a canonical estimator. The benchmark does not ask models to recover an ideal population parameter directly. In finite samples, even a textbook estimator may differ from the DGP parameter. We therefore compare model output to a canonical estimator applied to the same realised

Table 2: Benchmark controls. Each component supports a different claim and protects against a different failure mode.

Component	Implementation	Claim supported
Real context	259 papers; reconstructed RQ/DD/IC; consensus reference labels	Text-level method and direction agreement
Executable truth	100 fixed-seed DGPs; realised CSVs; canonical estimators	Numerical coefficient correctness
Role separation	Separate stages for construction, response, labeling, extraction, calibration	Reduces circular grading
Leakage control	Method-name blacklist and prompt leakage audit	Prevents trivial label recovery
Scorer audit	Coefficient extraction judge plus human validation audit	Reduces brittle post-processing errors
Claim boundaries	Exp A = agreement; Exp B = execution; calibration = confidence alignment	Prevents overclaiming

dataset. For scalar extracted estimate $\hat{\beta}_m$ and realised-data canonical estimate $\hat{\beta}_c$, L2b+ passes when

$$\frac{|\hat{\beta}_m - \hat{\beta}_c|}{|\hat{\beta}_c|} \leq \tau,$$

with default tolerance $\tau = 0.50$. Scenarios with $|\hat{\beta}_c| < 10^{-9}$ are marked unscoreable rather than divided. For Event Study scenarios, accepted post-event windows and scale conversions are frozen before scoring; valid cumulative post-event estimates are accepted only when they can be deterministically converted to the canonical per-day abnormal-return scale. This mapping is fixed before model comparison.

Why L3/L4 are supporting metrics. Exp A institutional context necessarily describes identifying variation, so method family can often be inferred by a trained reader. We therefore do not call L3/L4 method-blind detection or causal correctness. They are useful diagnostics of how models parse real research contexts, but the load-bearing correctness evidence comes from Exp B.

Scorer validation. A preliminary 30-case audit of L2b-pass/L2b+-fail examples found that regex extraction produced many scorer-side failures. The final scorer therefore uses a separate coefficient-extraction judge to identify the reported treatment-effect estimate from executed stdout, followed by deterministic numerical comparison against the canonical estimate. The extraction judge sees the executed code, stdout, scenario context, and method family, but not the evaluated model identity, canonical estimate, or L2b+ label. A completed 50-cell blinded human audit of this extraction instrument finds 90.9% numeric agreement among human-scoreable effect-present cells and 88.6% agreement on the induced L2b+ pass/fail decision; Appendix E gives denominators and disagreement cases. The judge can still err, so we treat L2b+ as an audited execution-grounded metric rather than an infallible oracle.

Threshold choice. The default L2b+ tolerance is relative error at most 50%. This threshold is intentionally broad: it allows finite-sample and implementation variation while still rejecting estimates that are off by a factor large enough to change practical interpretation. The binary threshold is a coarse workflow-success endpoint; the relative-error profiles and tolerance sweep test whether conclusions depend on a single cutoff.

5 Results

The results follow the same separation as the benchmark design. Exp A first shows what real-paper text agreement can and cannot tell us. Exp B then tests the executable claim: whether the model-written workflow recovers the canonical estimate. Finally, calibration asks whether the models themselves can identify when that workflow is wrong.

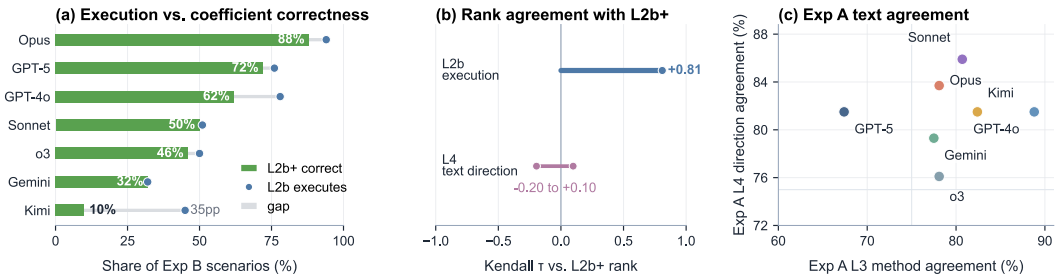


Figure 2: Headline result. **(a)** L2b execution and L2b+ coefficient correctness are distinct. **(b)** L2b ranking tracks L2b+ correctness (Kendall $\tau = +0.81$), while L4 text-direction scoring does not ($\tau \in [-0.20, +0.10]$). **(c)** Exp A L3/L4 agreement across the seven primary models; L3/L4 are diagnostics, not verified causal correctness.

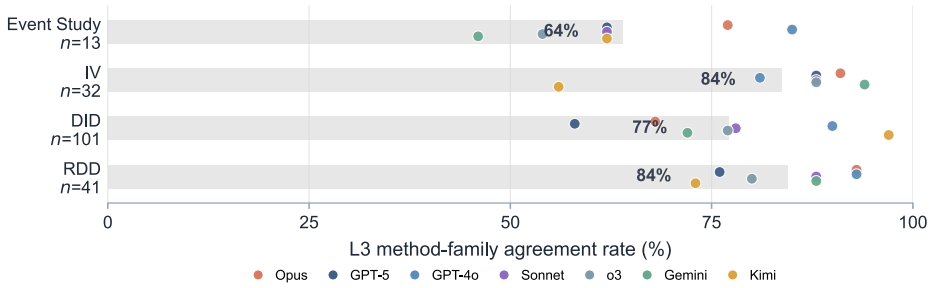


Figure 3: Exp A L3 method-family agreement by method family. Bars show cross-model means and colored markers show individual model scores. Event Study has the lowest mean agreement (64%, $n = 13$), while IV and RDD are higher despite lower corpus frequency. L3 is a text-agreement diagnostic, not verified causal correctness.

Finding 1: real-paper text agreement is useful but not decisive. On Exp A, L3 is scoreable for 187/259 papers and L4 for 92/259 papers. GPT-4o leads L3 (88.8%), Claude Sonnet leads L4 (85.9%), and GPT-5 is lowest on L3 (67.4%) but middle of the pack on L4 (81.5%). The rankings do not collapse to one textual ability score. Figure 2(c) summarizes the model-level separation between L3 method-family agreement and L4 direction agreement, while Figure 3 breaks L3 down by method family. Event Study has the lowest cross-model L3 mean even though it is not the most frequent method family. The 30-paper blinded human ambiguity audit reinforces this interpretation: agreement with the 4-LLM consensus is 60.0% for method family and 47.6% for direction. Exp A is therefore best read as a real-context text diagnostic, not as the paper's primary correctness endpoint.

Finding 2: runnable code is not necessarily correct. Figure 2 shows that L2b execution and L2b+ correctness separate the model panel. Final ES-aware canonical L2b+ rates span 10–88% across the seven primary commercial models: Opus 88%, GPT-5 72%, GPT-4o 62%, Sonnet 50%, o3 46%, Gemini 32%, and Kimi 10%. This spread is invisible if evaluation stops at whether the model wrote code.

Finding 3: execution ranking tracks numerical correctness. Across the seven models, the L2b ranking strongly matches the final L2b+ ranking (Kendall $\tau = +0.81$, Spearman $\rho = +0.93$). By contrast, L4 text-direction scoring has weak or negative association with L2b+ correctness ($\tau \in [-0.20, +0.10]$ across two frozen scorer variants). Execution is not sufficient for correctness, but it is a much stronger signal than text-level directional agreement. Because L2b+ is conditional on execution, Appendix C reports $P(\text{L2b+} \mid \text{L2b})$ to separate execution failures from executed-but-wrong estimates.

Finding 4: L2b+ rankings are not threshold artifacts. The L2b+ relative-error distributions show that stronger models concentrate executed estimates near zero, while weaker models accumulate large-error tails. Thus the ranking is better understood as a distributional difference in numerical

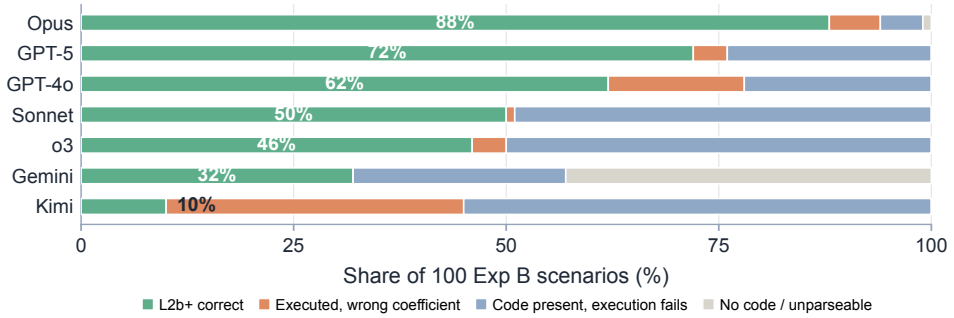


Figure 4: Execution cascade on Exp B. Bars partition 100 scenarios per model into L2b+ correct estimates, executed-but-wrong coefficients, execution failures, and missing or unparseable code. The orange segment shows why runnable code alone is not a sufficient endpoint.

Table 3: Calibration summary against final L2b+ v2 correctness. Gap is mean numerical confidence on correct outputs minus mean confidence on incorrect outputs. Small gaps indicate weak triage value.

Model	n	L2b+ v2	Conf. gap
Opus	100	88.0	+0.075
GPT-5	99	71.7	-0.045
GPT-4o	100	62.0	+0.016
Sonnet	99	49.5	-0.011
o3	100	46.0	-0.010
Gemini	48	31.2	+0.234
Kimi	100	10.0	+0.027

accuracy rather than an artifact of the default 50% cutoff. Appendix F reports the full relative-error profiles and tolerance sweep.

Finding 5: self-confidence does not identify mistakes. Calibration asks each model to rate its own previous Exp B output without seeing ground truth. Actual L2b+ rates vary sharply, but confidence gaps remain small for the main models. GPT-5 illustrates the asymmetry: it is second on L2b+ but has a slightly negative confidence gap. The self-assessment signal therefore does not provide a reliable deployment triage signal. Appendix F reports reliability diagrams for the retrospective confidence protocol.

Robustness check: the execution–correctness gap reproduces for an open-weights model. As a single cross-vendor check, we evaluate Llama-3.3-70B-Instruct on the same 100 Exp B scenarios using the same prompt and scoring pipeline. It reaches L2b $= 41\%$ and final ES-aware L2b+ $= 20\%$, placing it between Kimi and Gemini and reproducing the same qualitative gap between runnable code and numerically correct code. We treat this as a robustness check, not a complete benchmark of the open-source ecosystem.

6 Reproducibility and Human Audits

The final submission build is tied to a frozen audit state. The released artifacts include reconstructed Exp A paper fields, four-LLM consensus reference labels, 1813 Exp A outputs, 700 primary Exp B model-scenario execution records, 100 realised CSV datasets with canonical-estimator outputs, 646 calibration records, prompt hashes, scorer decision records, and cached model responses. Deterministic scoring can be rerun without new LLM calls; because commercial APIs are not bit-reproducible, the release treats cached JSON outputs as the reproducible model-output layer. Appendix A lists artifact paths and checksums.

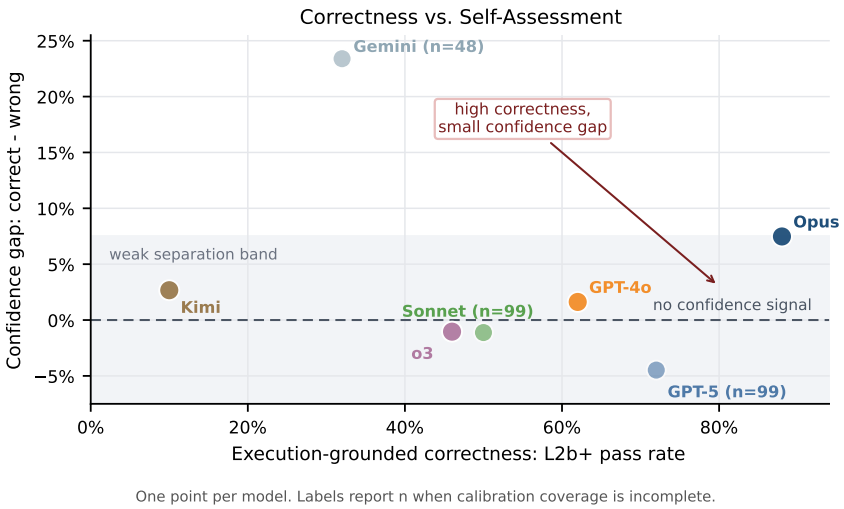


Figure 5: Correctness versus self-assessment. Each point is one model. The horizontal axis is execution-grounded L2b+ correctness; the vertical axis is the mean numerical-confidence gap between correct and incorrect workflows. High L2b+ accuracy does not imply a reliable confidence signal.

Human annotation is intentionally an ambiguity audit rather than a training or tuning input. A blinded 30-paper Exp A slice was labeled from the original PDFs after the automated benchmark freeze, without access to Phase-2 LLM votes, consensus labels, or model outputs. These human labels are not used to change the frozen consensus labels, prompts, model outputs, or scoring thresholds. Agreement with the four-LLM consensus is 60.0% for method family ($18/30$; Cohen's $\kappa = 0.606$) and 47.6% for direction ($10/21$; Cohen's $\kappa = 0.294$ among direction-scoreable cases). This supports our interpretation of Exp A as a real-context agreement diagnostic with substantial ambiguity, while Exp B supplies executable correctness.

A separate blinded human audit validates the coefficient-extraction instrument used for L2b+. The completed 50-cell audit finds 90.9% numeric agreement among human-scoreable effect-present cells and 88.6% agreement on the induced L2b+ pass/fail decision; Appendix E gives denominators and disagreement cases. This audit validates the measurement instrument but does not change the frozen headline L2b+ rates.

7 Limitations

First, Exp A labels are reference labels rather than causal ground truth. The final labels come from a four-LLM consensus after the final paper-field reconstruction pass, and 100 disagreement cases remain flagged for potential human adjudication. The completed blinded 30-paper ambiguity audit agrees with the consensus on 60.0% of method labels and 47.6% of direction labels, indicating that real-paper method and direction labels are noisy even under careful reading. This audit is useful as an ambiguity check, but it does not make the 259-paper Exp A corpus human-gold labeled. We therefore use Exp A to study text-level agreement in real research contexts, not to certify causal correctness. Because Exp A uses published papers, we also cannot fully rule out model familiarity with some studies from pretraining or public metadata.

Second, Exp B is synthetic and structured. This is necessary for executable ground truth, but it measures whether models can implement textbook workflows on known CSV data, not whether they can conduct an entire empirical research project. The DGPs cover four major method families but cannot represent all design choices in applied economics and finance. In particular, Exp B does not test literature review, data cleaning, design selection from ambiguous field constraints, or iterative robustness analysis.

Third, calibration is retrospective and prompt-specific. The protocol tests whether models can assess their own prior outputs under our confidence-elicitation prompt, not whether all possible uncertainty

interfaces would fail. Calibration coverage is also incomplete for some models. The negative result should be read as evidence that naive retrospective self-confidence is not enough, not as proof that no calibrated interface can be designed.

Fourth, the evaluated model panel is limited. The primary panel consists of closed or hosted frontier systems, which is appropriate for the deployment question that motivates the paper but does not characterize the full open-source ecosystem. We add one open-weights model as a cross-vendor robustness check; a complete benchmark of multiple Llama, Qwen, Mistral, DeepSeek, and domain-specific systems is left for future work. Model-level rank correlations should also be read as descriptive diagnostics over this evaluated panel, not population-level estimates over all possible LLMs.

Fifth, L2b+ is an audited measurement instrument rather than an oracle. It depends on the canonical specification, the relative-error tolerance, event-study scale conversions, and coefficient extraction from executed stdout. We mitigate these risks through frozen scorer rules, decision records, and a completed coefficient-extraction human audit, but individual-cell mistakes remain possible and should be treated as part of the benchmark's measurement uncertainty.

Finally, the benchmark is intentionally single-shot and non-interactive. Models receive fixed prompts and data previews, and the benchmark scores the returned workflow rather than allowing extended human-in-the-loop debugging or iterative robustness analysis. This design makes evaluation reproducible and comparable across models, but it does not measure how a supervised analyst might use an LLM over multiple rounds to improve a causal study.

8 Conclusion

CAUSALVERIFY shows that the right unit of evaluation for LLM-assisted causal inference is the workflow. A model may produce plausible causal text and runnable code while still estimating the wrong causal quantity. By separating real-paper agreement, execution, coefficient recovery, and self-assessment, the benchmark exposes where such workflows fail. Across seven evaluated LLMs, final L2b+ correctness ranges from 10% to 88%, execution ranking tracks coefficient recovery much better than text-direction scoring, and retrospective confidence does not reliably identify wrong workflows. For causal inference, the question is therefore not whether an LLM can produce an analysis that looks correct, but whether the workflow can be executed, checked against a specified target, and accompanied by uncertainty signals that are themselves audited.

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A Reproducibility Details

Appendix roadmap. Appendix A lists frozen artifacts and checksums. Appendix B describes the 30-paper Exp A ambiguity audit. Appendix C reports Exp B robustness diagnostics: canonical-estimator targets, conditional $P(\text{L2b+} \mid \text{L2b})$, tolerance sweep, scorer evolution, and method-family breakdown. Appendix D contains the descriptive RID pilot. Appendix E reports the 50-cell coefficient-extraction human audit. Appendix F contains additional diagnostic figures moved from the main text.

The final submission build is tied to a complete audit state. The audit tag `v11-freeze-2026-04-29` preserves the full provenance trail, while `causalverify_neurips2026.tex` provides a submission-length rendering of the same claims. The final submission PDF checksum is documented in `audit/SUBMISSION_BUILD_SUMMARY.md`.

Table 4: Frozen artifacts used by the final submission build. The full audit state remains available for inspection; the submission paper is a shorter rendering of the same frozen claims.

Claim	Frozen artifact
Exp A outputs	`experiments/exp_a/outputs/`
Exp A scores	`experiments/exp_a/auto_scores.csv`
Exp B L2b+	`l2b_plus_scores_canonical_judge_v2.csv` (primary 7; Llama R6 retained)
Calibration	`calibration_scores.csv`, `calibration_summary_v2.json`
Human validation	`audit/human_gold/human_vs_llm_consensus.md`
Paper checksum	`audit/SUBMISSION_BUILD_SUMMARY.md`
Gate evidence	`audit/V11_ACCEPTANCE_GATES.md`

B Human Annotation Details

The repository contains a paper-native labeling protocol and a completed blinded 30-paper ambiguity audit slice. The annotator labeled method family and direction from the original PDFs without access to Phase-2 LLM votes, consensus labels, or model outputs. One target paper (`paper_159`) was excluded because the available PDF was appendix-only; `paper_17` was added as a blinded replacement from the unsealed sample manifest, which contains only paper ID, difficulty tier, and PDF path. These human labels are not used to silently change the frozen Exp A labels. Human adjudication was conducted after the automated benchmark freeze as an ambiguity audit, not as a training, prompting, or tuning input; the labels are not used to construct prompts, select model outputs, or tune scoring thresholds.

C Exp B Robustness Details

This appendix reports supporting diagnostics for Exp B. These tables are derived from `l2b_plus_scores_canonical_judge_v2.csv` and do not replace the frozen headline L2b+ rates in the main text.

Table 5: Canonical-estimator targets used for Exp B L2b+ scoring. "Model output target" is the quantity the model is expected to report; "canonical comparison target" is the realised-data quantity computed by the frozen scorer.

Method family	Model output target	Canonical comparison target
DID	treatment-effect coefficient	realised-data DID regression coefficient
Event Study	post-event abnormal return / effect	canonical per-day post-event scale
IV	2SLS treatment coefficient	realised-data 2SLS coefficient
RDD	local jump at cutoff	realised-data local-linear RDD estimate

Because L2b+ requires code execution, raw L2b–L2b+ association is partly structural. Table 6 therefore reports the conditional metric $P(\mathrm{L2b+} \mid \mathrm{L2b})$, which asks whether executed code computes the right coefficient.

Table 6: Conditional Exp B correctness among executed outputs. "Wrong after execution" is L2b count minus L2b+ v2 count.

Model	L2b	L2b+	Wrong after exec.	P(L2b+|L2b)	Gap
Opus	94	88	6	93.6%	6 pp
GPT-5	76	72	4	94.7%	4 pp
GPT-4o	78	62	16	79.5%	16 pp
Sonnet	51	50	1	98.0%	1 pp
o3	50	46	4	92.0%	4 pp
Gemini	32	32	0	100.0%	0 pp
Kimi	45	10	35	22.2%	35 pp

Table 7: Scorer evolution on Exp B. Regex extraction is retained as a legacy diagnostic; the final headline uses the ES-aware v2 score. Entries are pass rates in percentage points.

Model	Regex	Judge	ES-aware v2	v2–Judge
Opus	64	71	88	+17
GPT-5	29	59	72	+13
GPT-4o	28	50	62	+12
Sonnet	27	31	50	+19
o3	18	37	46	+9
Gemini	15	29	32	+3
Kimi	6	8	10	+2

Table 8: L2b+ v2 by model and method family. Each cell reports passes/available scenarios.

Model	DID	ES	IV	RDD
Opus	29/30=97%	19/24=79%	23/24=96%	17/22=77%
GPT-5	20/30=67%	17/24=71%	23/24=96%	12/22=55%
GPT-4o	25/30=83%	13/24=54%	15/24=62%	9/22=41%
Sonnet	20/30=67%	21/24=88%	3/24=12%	6/22=27%
o3	8/30=27%	12/24=50%	16/24=67%	10/22=45%
Gemini	14/30=47%	5/24=21%	13/24=54%	0/22=0%
Kimi	5/30=17%	4/24=17%	1/24=4%	0/22=0%

Tolerance sweep. The robustness audit also recomputes pass rates for relative-error tolerances {0.10, 0.25, 0.50, 0.75, 1.00} among executed rows with a defined `rel_error_v2`. The sweep is reported in `paper/tables/exp_b_tolerance_sweep_primary7.csv` and `audit/exp_b_robustness/tolerance_sweep_primary7.json`. The sweep is conditional on execution and therefore complements, rather than replaces, the paper's primary all-scenario L2b+ rates.

D RID Pilot Ablation

Experiment C is a small descriptive pilot that reruns a 10-paper subset under the RID pre-commitment protocol. It is not used for the headline benchmark claims, because $N = 10$ per model is too small for model-level inference. We include it only as an audit artifact showing how pre-commitment can change text-level L3/L4 outcomes in either direction.

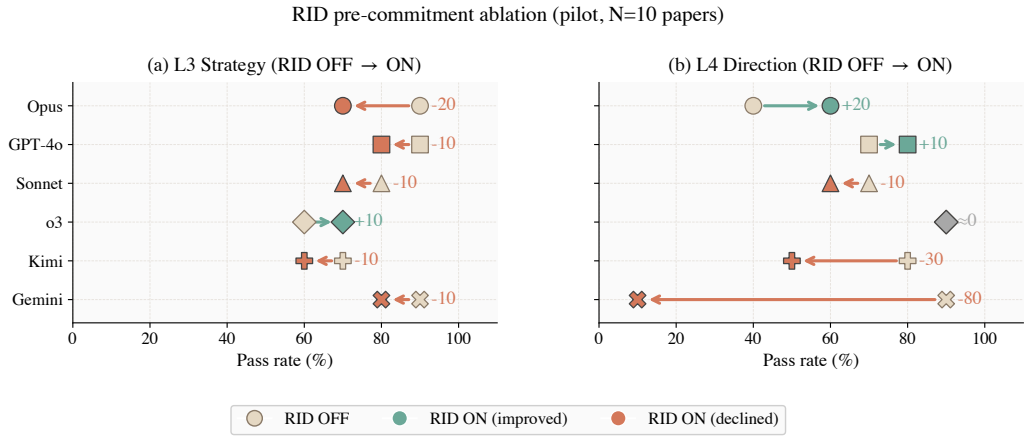


Figure 6: RID pre-commitment ablation pilot ($N = 10$ papers per model, descriptive only). Dots compare RID OFF and RID ON pass rates for L3 strategy and L4 direction. No individual comparison is statistically significant; the pilot is retained as an audit artifact rather than as evidence for the main claims.

E Coefficient-Extraction Human Audit

The coefficient-extraction judge is a measurement instrument: it identifies the scalar treatment-effect coefficient from executed R code/stdout before L2b+ compares that coefficient to the canonical estimator. The repository contains a blinded human-validation audit in `audit/l2b_judge_human_validation/`. The annotation form hides model identity, L2b+ label, canonical estimate, judge effect, regex effect, and ranking information. The completed audit covers 50 primary-panel L2b=1 cells sampled with seed 20260502: DID 13, Event Study 12, IV 11, and RDD 14. Among comparable audited cells, it found 90.9% numeric agreement and 88.6% agreement on the induced L2b+ pass/fail decision. Disagreements concentrate in event-study window choices and RDD sign/printing ambiguities. This audit validates the coefficient-extraction instrument; it does not change the frozen headline L2b+ rates.

Disagreement categories. Most disagreements arise from two sources: (i) Event Study outputs that report a cumulative post-event window rather than the canonical per-day scale, and (ii) RDD outputs where printed signs, jump orientation, or robustness estimates make the main effect ambiguous. These cases motivate treating L2b+ as an audited measurement instrument rather than an infallible oracle.

F Additional Diagnostics

This appendix collects supporting diagnostic figures for the main results. The main text reports the headline conclusions; the relative-error profiles, calibration reliability diagrams, and L4 scorer-operationalization diagnostic below support those conclusions but are not required for the central correctness argument.

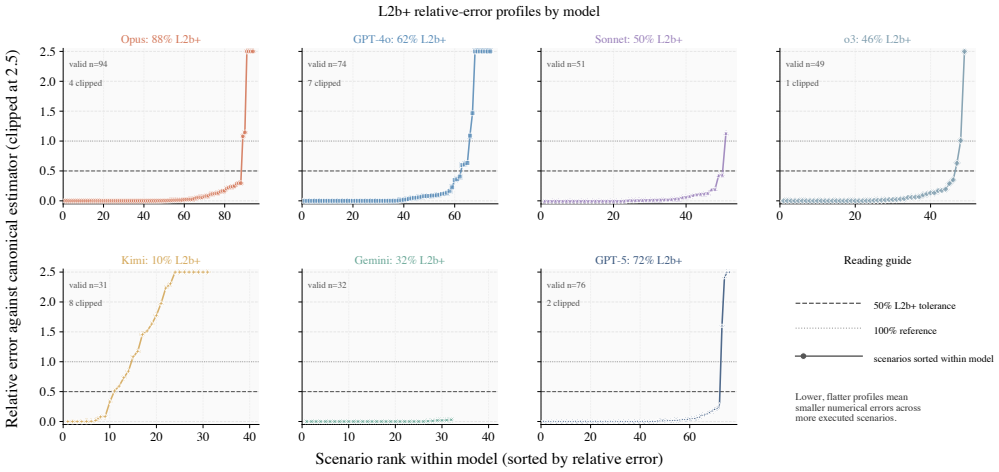


Figure 7: Additional Exp B robustness diagnostic. Each panel sorts executed estimates by relative error against the realised-data canonical estimator; the dashed line marks the default 50% tolerance. These profiles support the main-text claim that L2b+ rankings are not driven by a single cutoff.

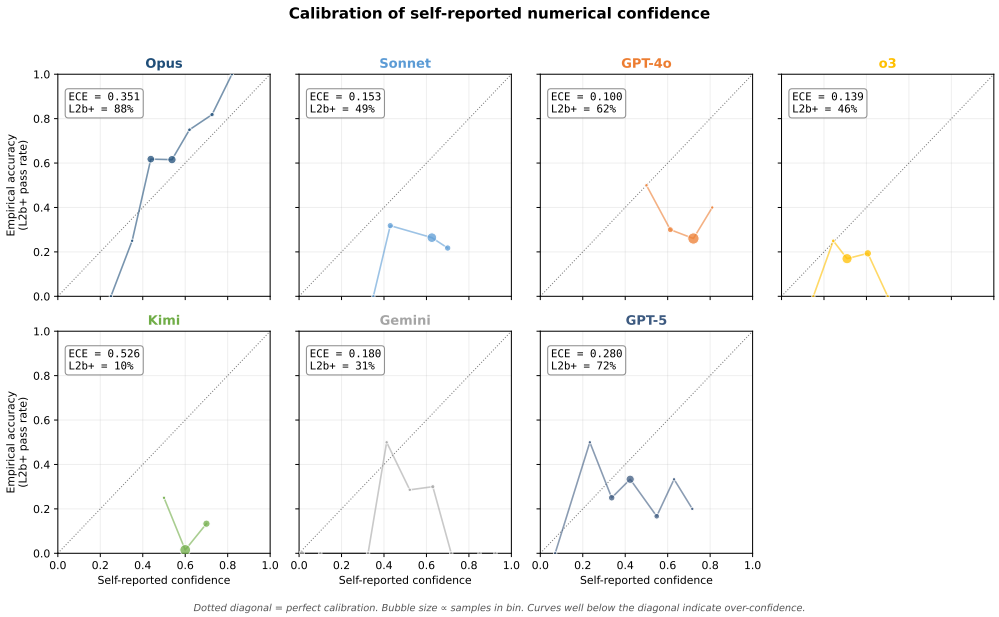


Figure 8: Additional calibration diagnostic. Reliability diagrams compare retrospective self-reported confidence with empirical L2b+ correctness. These diagnostics support the main-text finding that naive self-confidence does not reliably identify incorrect causal workflows.

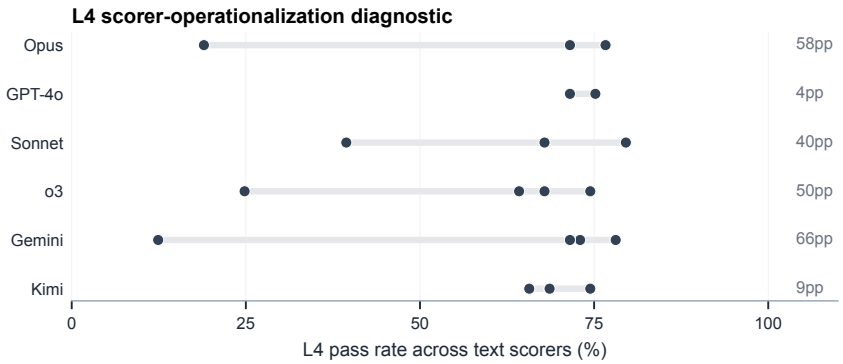


Figure 9: L4 scorer-operationalization diagnostic. Alternative frozen L4 text-direction scorers produce unstable pass rates across models. This diagnostic is not part of the primary seven-model headline figure because the scorer sweep was run on the six models with frozen scorer-variant outputs.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction define the scope as a benchmark for verifying LLM causal-inference workflows. They distinguish real-paper text agreement from execution-grounded coefficient correctness, and state that Exp A measures agreement diagnostics while Exp B provides the executable correctness endpoint.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Limitations section covers Exp A label ambiguity, the 30-paper human ambiguity audit, the fact that Exp A is not full human-gold ground truth, the synthetic scope of Exp B, the dependence of L2b+ on canonical estimators and coefficient extraction, model snapshot limits, and domain scope.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: [N/A]

   Justification: This is an empirical benchmark paper; no theoretical results or proofs are presented.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: [Yes]

   Justification: The anonymized release includes 259 active Exp A paper records, four-LLM consensus labels, 100 Exp B synthetic DGP scenarios with realised CSV data files and canonical-estimator targets, 1813 Exp A outputs, 700 Exp B primary execution cells, 646 calibration records, the L2b+ coefficient-extraction pipeline, prompt templates, scoring scripts, and frozen audit summaries.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient documentation and instructions for reproducing the main experimental results?

   Answer: [Yes]

   Justification: The anonymized repository submitted with the paper contains the code, frozen outputs, documentation, datasheet, component-level data-license notes, and reproduction commands. The README explains how to rerun deterministic scoring for Exp A, Exp B, and calibration without making new LLM API calls.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: No training is performed; this is an evaluation-only benchmark. The paper and release specify the evaluated models, prompt construction, Exp A consensus-label construction, Exp B DGP generation, realised-data canonical estimators, L2b/L2b+ scoring, and calibration protocol. Prompt templates and implementation details are included in the code release.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: The paper reports sample sizes for all scoreable subsets, model-level pass rates, rank correlations for the L2b/L2b+ and L4/L2b+ comparisons, and a relative-error distribution analysis showing that the main Exp B ranking is not an artifact of a single tolerance threshold. The paper does not make pairwise model-significance claims.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type, memory, time, etc.) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: The original model-generation runs use commercial LLM API calls and no local GPU compute. The deterministic scoring and figure-generation steps can be rerun from frozen outputs on a standard CPU environment. Token counts, costs, prompts, and run logs are preserved in the audit artifacts.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

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   Justification: The paper evaluates publicly available LLMs on published economics papers and synthetic scenarios. No human subjects are involved. The broader impact paragraph (Section 8) discusses risks of using LLMs for policy-facing causal analysis.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: [Yes]

    Justification: Section 8 (Broader impact paragraph) discusses the risk that LLMs used in policy-facing causal analysis without recognizing documented failure modes could support misleading conclusions. The benchmark itself is designed to surface these failure modes, which is a positive impact.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?

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13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets in a format that is easily accessible?

Answer: [Yes]

Justification: The benchmark is accompanied by a DATASHEET.md following Gebru et al. (2021), a detailed README with reproduction instructions, and structured JSON schemas for all papers and scenarios.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

    Answer: [N/A]

    Justification: No crowdsourcing or human subjects research was conducted. A 30-paper blinded author ambiguity audit and a 50-cell blinded coefficient-extraction audit were performed after the automated benchmark freeze; neither was used for training, prompt tuning, model selection, or threshold tuning. The labeling protocols and instructions are released in the repository.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether compensation was provided, etc.?

    Answer: [N/A]

    Justification: No human subjects research was conducted.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs in the research process?

    Answer: [Yes]

    Justification: LLMs are the subject of evaluation and are used in the benchmark pipeline for Exp A consensus labeling, field reconstruction, coefficient extraction, and calibration elicitation. The release includes prompts, model outputs, caches, and audit records needed to inspect those uses. The authors also used LLM assistance during coding and manuscript editing, with final responsibility for claims and text retained by the authors.